TachyHealth

AiAduit: Fraud, Waste and Abuse Engine

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Riyadh - Dubai - Delaware

TachyHealth

A clinically-led Organization

TachyHealth leverages AI, automation, and data science to enhance value-based healthcare. We optimize the medical journey with advanced systems for medical necessities, coding, analysis, and cost savings. Our deep technology solutions are considered best-in-class in the view of our partners and clients.

*Published Arxiv paper

250M+ Claims' activities processed and trained on our systems, majority from Saudi Arabia

SAR 10B+ Worth of claims processed through our systems

3,150+ Embedded medical guidelines including those of CHI

12x Our fine tuned LLM, gazal.ai , surpassed larger models by 12 folds*

80%+ Of our investment partners are from Saudi including leading players

We understand the Fraud, waste and abuse (FWA) multifaceted challenges

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Adverse impact on health systems

Diverts 2-25%* of valuable human and financial resources away from those who need them most, distorting the cost of care

Limited knowledge & insights from data

Limited ability to analyze millions of claims using traditional methods leads to untapped behavioral interventions & intelligence

Intensive and rigid operations

Manual interventions based on rigid rules give limited insights, allow the behavior of abusers to dominate the market practice

* JAMA Internal Medicine. 174 (7): 1067–1076 , JAMA. 2019;322(15):1501-1509, TachyHealth Analysis

Taming the healthcare FWA beast: AI solutions for value-driven outcomes

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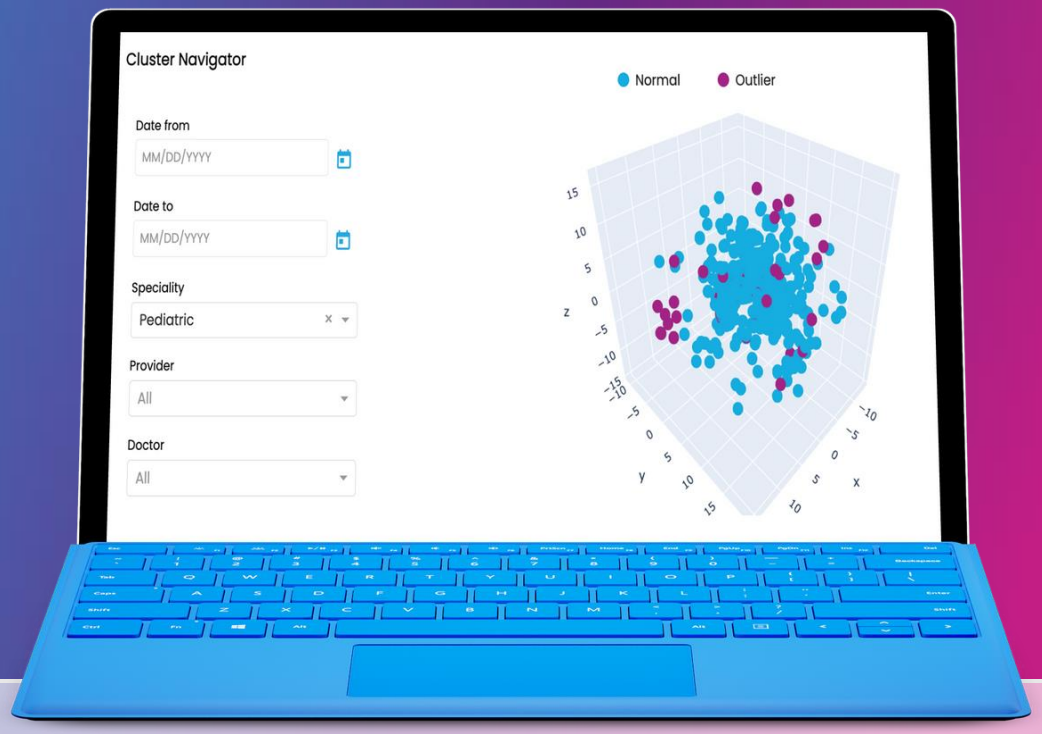
Ai Audit

Our FWA flagship solution, **AiAudit**, harnesses the power of **artificial intelligence and data science** in the medical claims and clinical encounters, enabling **large-scale adjudication, audits and reviews** with enhanced efficiency. This promotes real-time, data-driven fraud, waste, and abuse (FWA) detection, ultimately increasing team productivity, optimizing performance, and offering a gamified user experience.



Increase health quality and patient safety

Improve the financial sustainability & value-based care

Enhance the efficiency and optimize the performance

FWA key analytical indicators

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Ai Audit

- Billing and financial patterns for providers & doctors
- Outlier and anomaly clustering algorithms
- Similarity-based (over-cooking) behaviors
- Suspicious relationships between parties
- Medical and clinical encounter performance
- Time-series and sequence-based analysis

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The process to handle data on AiReview

Our AI-powered system goes through three stages to deliver decisions on whether a claim or approval is an outlier or not alongside features for investigation

Pre-process

The raw data undergoes an Extract, Transform, and Load (ETL) pipeline to prepare the data for subsequent stages and apply machine learning pipeline

Hybrid model

AiReview employs a hybrid approach, utilizing both machine learning and rule-based edits, to deliver optimal results with our proprietary method

Interpretation

Each activity & claim is processed to generate decisions using a scoring and probabilities. Features that are worthy of investigation are incorporated to provide explainable AI

Scaling FWA with a gamified experience using our hybrid model

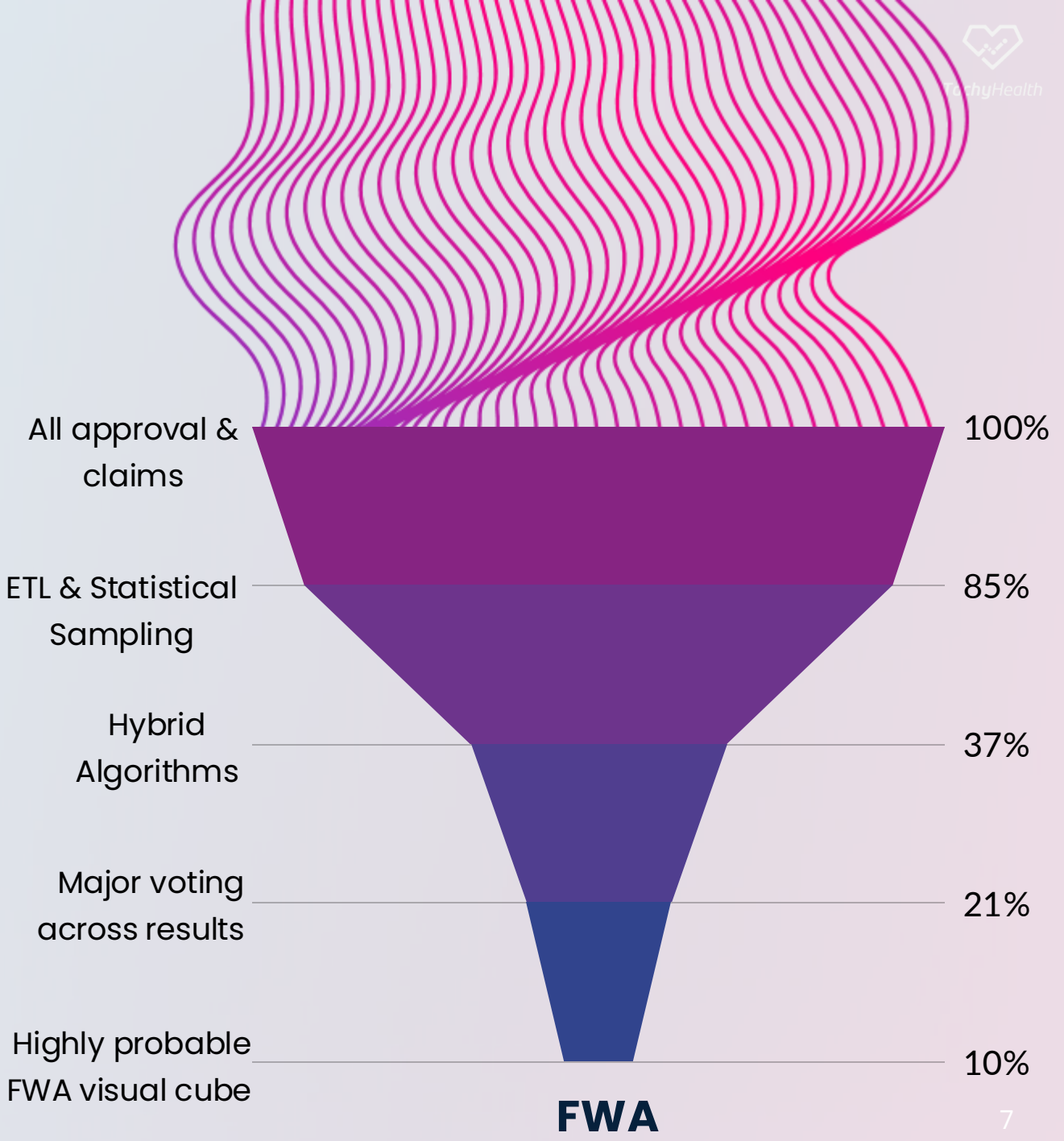
Audit team can detect outliers and abnormalities through an easy visual representation using AI in the loop to maximize audit efficiency. Our pipeline takes all data at intervals and provides reports in real-time.

Activities FWA

Provider FWA

Clinician FWA

Members FWA



AllFWA activities from different models are processed across 4 models

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Suspicious member behavior

Although this happens less frequently, the engine identifies patterns, discovers anomalies and trends especially shopping around and using cards with uncovered family members

Suspicious provider behaviour

Detect the FWA behaviors from providers who are consistently voted as suspicious based on the hybrid model which takes into consideration benchmarking based on similarity

Suspicious clinician behaviour

Detect potentially suspicious patterns from doctors based on clustering them across specialties, facilities, levels of expertise, and other factors. Scorecard is represented accordingly

Suspicious activities

Our most powerful unsupervised learning model that comprehend every single data point and identify patterns of outliers accordingly

Interactive Dashboard for Investigating Outliers by Insurance Companies

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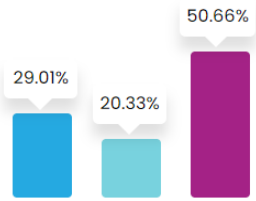
Total claims value
27,018,051.55 SAR

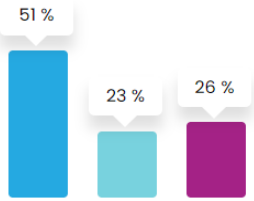
- 7,838,477.2 SAR Clean claims
- 5,492,286.9 SAR One outlier in claims
- 13,687,287.45 SAR Multiple outliers in claims



Total number of claims
38,152

View Report

- 19,531 Clean claims
- 8,680 One outlier in claims
- 9,941 Multiple outliers in claims



ML Models Insights

Total claims
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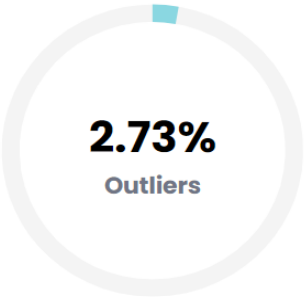
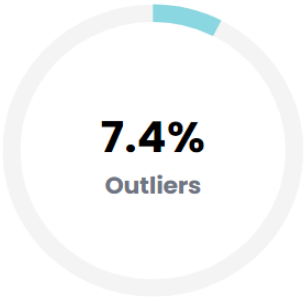
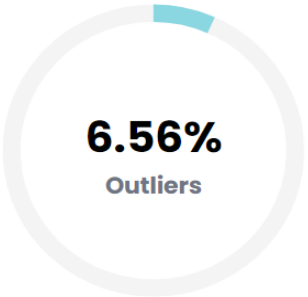


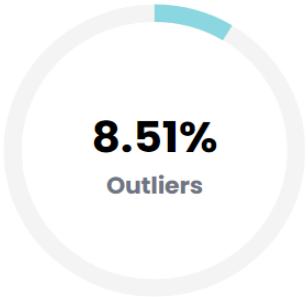
Total Providers
622



Total Doctors
5,884



Total Patients
3,607



Activities FWA | Provider FWA | Clinician FWA | Members FWA

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Thank You

For further information, contact us.

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